

SRIJAN PABBI

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EDUCATION

University of Waterloo, ON, Canada

Sept 2021 - Aug 2023

M.A.Sc, Master of Applied Science, Electrical and Computer Engineering

Percentage: 95.80%

On Statistical Variability Analysis of Anvil (Counterfeit Detection System)

Thesis Advisor: Prof. Sebastian Fischmeister

Coursework:

- ECE 606 - Algorithm Design and Analysis
Skills Gained: Discrete Math and Proofs, Data Structures, Asymptotic Analysis, Design Techniques, Non-determinism, and NP reductions.
- ECE 650 - Methods and Tools for Software Engineering
Skills Gained: C++, Python object-oriented programming, data structures, parallel programming, regular expressions, multi-processing, concurrency.
- ECE 657A - Data & Knowledge Modelling & Analysis (Machine Learning)
Skills Gained: Data Pre-processing, Feature Engineering, Statistical Estimation, Representation Learning, Classification Techniques, Clustering and Un/Semi-supervised Techniques, Deep Learning (ANN, CNN, RNN, Autoencoders)
- MSCI 718 - Statistical Methods for Data Analytics (Intro to Statistical Learning)
Skills Gained: Statistical Hypothesis Testing, Estimation in depth, Bias Variance Trade-off, Bayesian Approach, Multiple Linear Regression and Residual Analysis
- CS 658 - Computer Security and Privacy
Skills Gained (Overview and Hands on): Program and OS Security, Network Security, Internet and Web Security, Data Security and Privacy, Differential Privacy.

University of Delhi, New Delhi, India

Aug 2015 - July 2019

Netaji Subhas Institute of Technology (NSIT),

CGPA: 8.525

{Upgraded to Netaji Subhas University of Technology, w.e.f. 09/26/2018}

Percentage: 77.75%

Bachelor of Engineering, Electronics and Communication Engineering

Coursework: Computer Architecture, Microprocessors, Operating Systems, Software Engineering, C/C++ Programming, Digital Signal Processing, Control Engineering, Probability Theory, Digital Circuit Design, Analog Circuit Design, Network Analysis.

RESEARCH AND WORK EXPERIENCE

Graduate Researcher

University of Waterloo, Department of Electrical and Computer Engineering. Thesis Advisor: Prof. Sebastian Fischmeister

Sept 2021 - Present

Working on Physics Based Cyber-Physical Security [Product Design and Refinement for Palitronica Inc.]:

- Design of (Anvil) measuring instrument to be used for verification of hardware targets (semiconductor ICs), separating original from counterfeits.
- Statistical Variability Analysis of Measurement System: Evaluating and modelling different sources of variability and devising procedures for calibration and long-term repeatability and reproducibility of results.

Research Engineer

Centre for Development of Telematics (C-DOT), Govt. of India

July 2019 – Aug 2021

As part Optical Core Networks Group, I worked on development of optical line cards for CDOT's DWDM System [Product Design and Testing].

- Circuit Schematic Design and supervise PCB CAD with high-speed signal integrity constraints.
- Design the architecture for Data and Control Path for SoC

INTERNSHIPS

Software R&D Intern

Mentor Graphics Noida, Uttar Pradesh, India

Dec 2018 - Feb 2019

Developed a signal and exit handler, annotation infrastructure for software messages and file/data encryption infrastructure in C++ for Symphony, Mentor Graphics' latest High-Performance Mixed-Signal Platform

Project Trainee, Internship

Texas Instruments Inc. Bangalore, Karnataka, India

May - July 2018

Theoretical & Experimental Analysis of Class-D (LSR) Output Filtering Circuitry: In Low Power Audio Amplifiers Team of TI on a research project for performance benchmarking of the audio amplifier ICs.

GENERAL OVERVIEW OF SKILL SET

Programming: Experienced at C/C++ and Python, Familiar with Java, C#, R.

ML Experience: Data preparation and feature engineering, representation learning, classification algorithms, and deep learning (TensorFlow)

Other: Design of Experiments (DOE), Embedded Hardware and Software Design, Verilog, VHDL, Digital & Analog Circuits Design, Git, PCB CAD.

ACADEMIC PROJECTS

Survey Paper on Security, Privacy and Safety Concerns of Connected and Autonomous Vehicle (CAV) Systems (CS 658) **Winter 2022**

Part of course requirement, did a research survey paper on potential cyber security vulnerabilities and risk areas in existing and upcoming connected and autonomous vehicle systems and suggested defensive and preventative counter measures/deterrents to such risks.

Voter Tendency Prediction based on Marital Status and State of Residence (MSCI 641): **Winter 2022**

Using the techniques learned in MSCI 641, built a logistic regression model for predicting voter intention using the Pew Research Centre pre polls data for 2008 US Presidential election campaign and 2008 election result. Applied Logistic regression with partially pooled intercepts and slopes for the predictors.

Open-Source Scientific Instrumentation: Undergrad B.E. Project. **Feb - May 2019**

Built a PCR Thermo-cycler for DNA and nucleic acid amplification (used in every DNA based analysis) with a Wireless Biochemical Reaction Multi-meter (pH and temperature) with data tracking, storage and monitoring with Android app.

Mathematical Alarm Clock (On 8085 using Assembly): **Jan - May 2018**

Designed an 8085-microprocessor based Alarm clock which prompts a mathematical question to shut off the alarm.

Talkative Tom - Hardware & Embedded Software Implementation: **Aug 2016 - Jul 2017**

Design, fabrication, and testing of an audio storing & playback device with pitch modification using MSP-430 TI-microcontroller / Atmega328P microcontroller, SPI Serial Flash for Storage & class-D amplifier for speaker

TEACHING, MENTORING & VOLUNTEER EXPERIENCE

Teaching Assistant **University of Waterloo, Department of Electrical and Computer Engineering**

- ECE 250 - Algorithms and Data Structures: Guiding students to build course projects, held weekly interactive sessions, and graded assignments and exams **Winter 2023**
- ECE 198 - Project Studio: Responsible for mentoring 30 groups of over 60 students to conceive, plan, build and present/demo a functioning engineering project aimed towards solving a problem in domain of climate change. **Fall 2022**

Student Contributor **Centre for Electronics Design and Technology (CEDT) NSIT** **June 2016 – May 2019**

CEDT is a "Maker Space/Open Access Lab" at NSIT. Our work here includes making academic projects, pedagogy focused tools, mentor students, conduct workshops and trainings within and outside NSIT.

General Secretary **Junoon The Photography Club of NSIT, New Delhi, India** **Jan 2016 - May 2019**

Leading cultural fest covering teams every year, organising online photography competitions, workshops, finding sponsors, networking with people and other photographers.

Instructor & Mentor

- **Tinkering Workshop for Army Goodwill Schools at IIT Jammu:** Workshop children studying in remote schools, introducing the idea of tinkering and DIY. Demonstrated and trained students on simple physics-based electronics experiments and projects. **Jan 2018**
- **Arduinodaya Workshop, NSIT:** Workshop on Embedded Systems design using the pedagogy project pcb board Arduinodaya designed by us. **Jan 2018**
- **TI-CEPD Internship Program NSIT and Chitkara University:** A month long TI micro-controller based embedded system design training program at NSIT Campus and a second month at Chitkara Campus. **June & July 2017**

INTERESTS AND HOBBIES

Nature, Portrait and Food Photography, Swimming, Skiing, and Cooking.